A Study on Determinants of Foreign Exchange Rates in Bangladesh

Abstract

This paper aims to identify the principal determinants of the foreign exchange rate to support informed decision-making for the effective management of this economic phenomenon in Bangladesh. Five independent variables were initially selected for investigating their relationship with the exchange rate. The preliminary step in constructing a multiple linear regression model entails assessing multicollinearity among the independent variables through the application of the Variance Inflation Factor (VIF) and tolerance. After completing the regression analysis, statistical significance was evaluated through F-tests and t-tests to ascertain their effect on the dependent variable. The data compilation spanned a period of twenty years, from 2005 to 2024. The variables, specifically interest rate, external debt, and remittance inflows, were ultimately used, and the study determined that these factors have a significant influence on the exchange rate. Furthermore, the model has the ability of incorporating the foreign exchange rate phenomenon with an explanatory power of 95.8%.

Keywords: Foreign exchange rate, Interest rate, External debt, Remittance inflows, Multiple linear regression.

1. Introduction

The foreign exchange rate is a vital indicator of a country's macroeconomic health, reflecting its trade competitiveness and currency stability. It signifies the rate at which one currency is exchanged for another, functioning as an indicator of economic stability. For economies heavily reliant on trade, such as Bangladesh, maintaining exchange rate stability is essential, as it impacts import expenses, export income, inflation rates, and foreign reserve levels. Changes in the local currency, the Taka, relative to the US Dollar have extensive implications for the economy. This volatility influences national savings, the investment environment, and the purchasing power of individuals, rendering it a matter of considerable importance beyond mere theoretical considerations. Therefore, maintaining stability in the exchange rate is vital for the overall health and development of the economy.

In the context of Bangladesh, the significance of the exchange rate is profound, given the nation's requirement to import substantial quantities of food, raw materials, and technological goods annually. Since Bangladesh switched from a fixed to a floating exchange rate system in the 2000s, market mechanisms and macroeconomic fundamentals have interacted more. Even small currency swings can affect fiscal policies and investor decisions. Today, market forces decide Bangladesh's

exchange rate, but the Central Bank still oversees it. Without precise exchange rate forecasts and analysis, foreign exchange reserves are nearly unsustainable. Thus, Bangladesh's exchange rate movements and factors are the focus of this study.

The foreign exchange rate of Bangladesh is predominantly affected by various particular macroeconomic variables. The primary concentration of this research is on interest rates, external debt, and remittance inflows. This research utilises statistical analysis to illustrate the impact of these three components on Bangladesh's exchange rate over recent years. Owing to the escalating impact on foreign exchange reserves and the rising demand for the dollar in the global market, these factors have become more significant than ever. According to this context, the present report has been developed with the aim of examining the influence of the primary factors affecting the foreign exchange rate in Bangladesh.

2. Literature Review

A significant amount of empirical and theoretical research has investigated the dynamics of exchange rates and their relationship with essential macroeconomic variables, considering the pivotal role of the exchange rate in influencing macroeconomic stability, external competitiveness, and financial sustainability in developing economies. The article examines relevant domestic and foreign literature to build the theoretical foundation of the study, synthesise current empirical findings, and highlight unsolved issues within the literature. This review critically assesses previous research, revealing methodological deficiencies and empirical discrepancies that underscore the need for the current study on the determinants of the Foreign Exchange Rate in Bangladesh.

Empirical investigations into the exchange rate in Bangladesh remain comparatively scarce, although multiple studies have produced significant insights into its macroeconomic determinants (Chowdhury & Hossain, 2014; Hasan & Islam, 2023). Bristy (2017) recognized broad money supply, GDP, gross capital formation, primary income payments, foreign debt stock, and gross national income as key determinants of the exchange rate. Hasan and Islam (2023) additionally reported that variables such as the current account balance, foreign exchange reserves, and deposit rates demonstrated moderate explanatory power, whereas inflation, lending rates, FDI, and net financial accounts exhibited comparatively weaker influence.

Kamal Uddin et al. (2013) demonstrated that an increase in the money supply and escalating debt-servicing obligations contribute to the depreciation of the Bangladeshi Taka, whereas the accumulation of foreign reserves exerts an appreciating influence (Yousuf et al., 2020). Khan (2021) endorsed this perspective by asserting that currency depreciation has the potential to boost exports and promote long-term economic development. Chowdhury et al. (2014) recognized exchange rates, remittances, domestic interest rates, broad money, and GDP as key factors influencing foreign exchange reserves (Naima et al., 2025). Islam (2009) analyzed reserve

adequacy and management, emphasizing internationally recognized benchmarks, while Alam and Rahim (2013) affirmed that reserve dynamics substantially impact other financial markets.

Hasan (2018) found a positive and statistically significant relationship between foreign exchange reserves, interest rates, and market capitalization in the Dhaka Stock Exchange, whereas Golder et al. (2020) observed a negative long-run association. Recognizing Bangladesh's heavy reliance on the US dollar for foreign trade, Khaleque et al. (2024) integrated NEER and REER movements with reserve trends to propose a comprehensive analytical framework.

Exchange rate fluctuations have also garnered scholarly interest. Alam and Rahman (2012), employing GARCH models, established that historical volatility exerts a substantial influence on current volatility. Abdullah et al. (2017) emphasized unresolved methodological challenges, including pronounced skewness and kurtosis in exchange rate returns. Khan and Ahmed (2024), utilizing the Classical Export Demand Function, demonstrated that depreciation of the exchange rate negatively impacts exports in both the short and long term. Kemal (2006) similarly determined that volatility promotes exports while inhibiting imports (Chowdhury & Hossain, 2014).

Recent research has highlighted the presence of nonlinear and asymmetric effects. Shahriar et al. (2024) observed that previous research in Bangladesh had largely neglected asymmetric effects of exchange rates, despite Jonaed (2019) highlighting negative long-term consequences. Hossain (2008) documented a significant positive correlation between foreign direct investment inflows and trade flows, while noting that previous research predominantly emphasized trend analysis rather than comprehensive econometric modelling. Together, these studies highlight Bangladesh's primary macroeconomic variables while emphasizing the necessity for more thorough and integrated exchange rate analysis.

The reviewed studies provide substantial empirical evidence regarding the influence of macroeconomic variables on exchange rate behavior, reserve accumulation, and trade dynamics in Bangladesh. Hasan and Islam (2023) identified GDP growth, remittances, foreign exchange reserves, and the current account balance as the most influential exchange rate determinants, with positive coefficients, although interest rates exhibited a negative correlation in preliminary analyses. Chowdhury and Hossain (2014) found GDP growth and the current account balance to be statistically significant, with an explanatory power of 76 percent.

Khan (2021) demonstrated that nominal exchange rate movements and FDI positively contribute to economic development, while external shocks such as pandemics adversely affect growth. Naima et al. (2025) reported exceptionally strong explanatory power of reserve dynamics, revealing both positive and negative associations with various macroeconomic indicators. Alam and Rahim (2013) confirmed the statistically significant impact of exchange rates on foreign exchange reserves, while Hasan and Chokroborty (2024) documented the influence of reserves, inflation, and call money rates on stock market capitalization.

Volatility-focused studies by Yousuf et al. (2020) and Abdullah et al. (2017) revealed persistent volatility clustering and leverage effects in the BDT–USD exchange rate. Khan and Ahmed (2024) and Shahriar et al. (2024) provided robust evidence of asymmetric and nonlinear effects of exchange rate fluctuations on exports. Hossain (2008) further established the positive contribution of FDI to balance-of-payments performance.

Despite comprehensive research from different perspectives, current studies predominantly concentrate on individual relationships, particular markets, or nonlinear dynamics. Consequently, a significant research absence persists concerning the concurrent analysis of multiple macroeconomic determinants within a comprehensive OLS framework, especially in the context of Bangladesh's exchange rate fluctuations. This research seeks to fill that gap.

3. Objective of the Study

This study aims to empirically investigate the macroeconomic determinants of the BDT/USD exchange rate in Bangladesh. By employing a multiple linear regression framework, the research quantifies the individual and collective impacts of key explanatory variables on exchange rate dynamics. The study further seeks to establish a robust analytical model for forecasting currency fluctuations, providing a statistical basis for assessing macroeconomic trends. Ultimately, the findings offer evidence-based insights to assist relevant authorities in formulating strategic interventions to ensure exchange rate stability and long-term macroeconomic resilience.

4. Variable Definition

For developing the multiple linear regression model, this report looks at one dependent variable and five independent variables. Here, the Foreign Exchange Rate (FER) is the dependent variable that varies in relation to changes in the independent variables. This model uses the following independent variables:

- Inflation Rate
- Interest Rate
- External Debt
- Remittance Inflows
- Trade Openness Index

4.1 Explanation of independent variables

- *Inflation Rate:* The inflation rate indicates the persistent rise in the overall price level of goods and services within an economy over a period of time. It directly impacts purchasing

capacity, the cost of living, and the actual value of currency. High inflation diminishes the purchasing power of a currency, rendering domestic products comparatively costlier than foreign goods, thereby undermining the currency's external valuation. In an economy such as Bangladesh that relies heavily on imports, fluctuations in inflation and exchange rates are intricately connected via the exchange rate pass-through effect, whereby currency depreciation elevates import costs and subsequently accelerates domestic inflation. Central banks address inflationary pressures by implementing contractionary monetary policy measures, typically through adjustments to interest rates. Although managing inflation contributes to currency stability, overly aggressive tightening could hinder investment and economic expansion. Thus, inflation interacts dynamically with trade, monetary policy, and exchange rate stability, generating feedback effects that influence macroeconomic outcomes.

- *Interest Rate:* The interest rate refers to the cost of borrowing or the return on lending money, usually expressed as a percentage over a specified period. It is a primary instrument of monetary policy used by central banks to influence inflation, investment, and consumption. Higher interest rates attract foreign capital inflows as investors seek better returns, increasing demand for domestic currency and strengthening its value. This mechanism operates through capital mobility and portfolio adjustments in global financial markets. However, if inflation rises faster than nominal interest rates, real returns decline, reducing currency attractiveness. Excessively high interest rates can also suppress domestic investment and economic activity. Therefore, the relationship between interest rates and exchange rates depends on inflation dynamics, investor confidence, and overall macroeconomic stability.

- *External Debt:* The portion of a country's total debt borrowed from non-resident lenders, including foreign governments and international financial institutions. It must be serviced in foreign currency, requiring sufficient export earnings or reserve accumulation. Rising external debt increases demand for foreign currency during debt servicing periods, placing depreciation pressure on the domestic currency. In developing economies, excessive external borrowing without productive investment weakens fiscal sustainability and exchange rate stability. High debt levels increase exposure to exchange rate risk and inflationary pressure, particularly when currency depreciation raises repayment costs. While external debt can finance development and infrastructure, mismanagement can lead to balance of payments stress, reduced creditworthiness, and long-term currency instability.

- *Remittance Inflows:* The income transfers made by migrant workers to their home countries represent a substantial source of foreign money. These inflows increase the supply of foreign currency in the local market, facilitating currency appreciation and enhancing balance of payments circumstances. In numerous emerging economies, remittances enhance consumption, alleviate poverty, and promote economic stability. Nevertheless, persistent inflows may result in real exchange rate appreciation, diminishing export

competitiveness due to the Dutch Disease phenomenon. The influence of remittances is contingent upon expenditure behaviors, exchange rate systems, and monetary policy reactions. Furthermore, the depreciation of exchange rates may stimulate increased remittance inflows, resulting in reverse causality and intricate adjustment processes.

- *Trade Openness Index:* The index measures the extent of a nation's integration into international trade by computing the ratio of total trade to gross domestic product (GDP). Increased trade openness signifies enhanced engagement in international markets and greater exposure to global economic conditions. Open economies generally exhibit more stability in real exchange rates owing to varied trade frameworks and effective adjustment mechanisms. Trade openness strengthens competitiveness, increases current account responsiveness to exchange rate fluctuations, and restricts local price-setting authority, leading to reduced inflation. Nonetheless, heightened transparency also amplifies vulnerability to external disturbances. The efficacy of stabilizing exchange rates is contingent upon supportive fiscal, monetary, and institutional policies.

5. Data & Methodology

The primary objective of this study is to develop an appropriate regression model to analyze the relationship between economic factors associated with Bangladesh's foreign exchange rate, and to effectively explain the behavior of the dependent variable in relation to the relevant explanatory variables. This study employs a quantitative and explanatory research approach. In this study, the Foreign Exchange Rate is treated as the dependent variable, while the other selected factors are incorporated as independent variables, as briefly discussed in the antecedent section. Given the nature and aims of the study, a multivariate regression analysis has been conducted.

In regression analysis, multiple techniques may be appropriate depending on the specific circumstances. Since all variables in this research are continuous and adequate for elucidating the dependent variable, the Multiple Linear Regression model has been selected as the most suitable choice. however, before developing the model, It is crucial to address and eradicate any possible overlaps or interdependencies among the independent variables. For this purpose, a multicollinearity assessment has been performed. The analysis was carried out through multiple stages, during which the Correlation Matrix was omitted due to the existence of numerous interdependent variables.

Following the multicollinearity test and prior to building the regression model, the suitability of the selected variables was ensured. Subsequently, F-tests and t-tests were applied to verify the statistical significance of the regression function and its coefficients. Simultaneously, the R^2 and Adjusted R^2 values were carefully observed to determine whether the independent variables accurately explain the dependent variable. All statistical analyses for the research were performed using IBM SPSS-27 software.

6. Data Source

All data used in this study are secondary data, collected from various internationally recognized and reliable sources. For the purpose of the research, annual data from 2005 to 2024 have been gathered. The World Bank database has been used as the primary data source, as it provides standardized and comparable economic data for developing nations. Additionally, data collected from the Federal Reserve Economic Data (FRED) database have been utilized, which are high-quality, regularly updated, and widely used in academic research. The information obtained from these sources ensures the reliability and validity of this study.

Table 01: Data source

Year	Y	X_1	X_2	X_3	X_4	X_5
2005	64.33	7.05	10.62	18503086016	6.21	34.40
2006	68.93	6.77	11.66	20160932717	7.56	38.11
2007	68.87	9.11	12.64	21523544177	8.24	39.94
2008	68.60	8.90	12.89	23346917708	9.76	42.62
2009	69.04	5.42	13.33	25377795050	10.27	40.09
2010	69.65	8.13	12.22	26571755072	9.41	37.80
2011	74.15	11.40	13.32	27046084648	9.39	47.42
2012	81.86	6.22	13.94	29155976742	10.59	48.11
2013	78.10	7.53	13.59	32454687017	9.24	46.30
2014	77.64	6.99	12.95	35271839355	8.67	44.51
2015	77.95	6.19	11.71	38704937144	7.84	42.09
2016	78.47	5.51	10.41	41569252898	5.12	31.33
2017	80.44	5.70	9.54	51068283465	4.60	30.00
2018	83.47	5.54	9.65	57125459321	4.84	32.51
2019	84.45	5.59	9.56	62468325029	5.23	31.58
2020	84.87	5.69	8.30	73550857817	5.82	26.27
2021	85.08	5.55	7.32	91477590072	5.33	27.72
2022	91.75	7.70	7.12	97020160351	4.67	33.78
2023	106.31	9.88	7.57	101447458740	5.07	30.98
2024	115.60	10.47	9.85	103638000000	6.04	26.78

Source: World Bank database and The Federal Reserve Economic Data (FRED)

y= Exchange Rate

X_1= Inflation Rate

X_2= Interest Rate

X_3= External Debt

X_4= Remittance Inflows

X_5= Trade Openness Index

7. Results and Discussion

7.1 Multicollinearity Test In order to ensure that the Multiple Linear Regression (MLR) model produces reliable results, it is necessary to perform a multicollinearity test to confirm that the independent variables do not exhibit any significant overlap with one another. The existence of overlapping variables may reduce the accuracy of the estimated coefficients and increase the complexity of interpreting the results. This diagnostic highlights the importance of examining distinct factors that individually influence the foreign exchange rate in Bangladesh. There are three widely recognized methods for detecting multicollinearity.

7.1.1 Correlation Matrix: We'll develop a Correlation Matrix to analyze independent variable correlations. When any two variables have an absolute correlation coefficient greater than 0.7, multicollinearity could exist. This criterion is beneficial, but further research is required to confirm results.

Table 02: Correlation Matrix

Variable	EXR	INF	IR	EXTD	REM	TOI
Exchange Rate	—					
Inflation Rate	0.264	—				
Interest Rate	-0.589**	0.165	—			
External Debt	0.898**	0.085	-0.845**	—		
Remittance Inflows	-0.509*	0.237	0.907**	-0.695**	—	
Trade Openness Index	-0.521*	0.231	0.873**	-0.712**	0.859**	—

* Correlation is significant at the 0.05 level (2-tailed).

The above Correlation matrix shows all correlation coefficients alongside their significance level.

Based on the correlation matrix data, a significant positive association (r = .898) is observed between Bangladesh's exchange rate and external debt, suggesting that an increase in debt

contributes to the depreciation of the Taka. Conversely, notable negative correlations are identified between the exchange rate and interest rates (r = -0.589), remittance inflows (r = -0.509), and the trade openness index (r = -0.521), indicating that increases in these factors contribute to the strengthening of the Taka. Notably, interest rates exhibit a strong positive correlation with remittances (r = .907) and trade. At the moment, the inflation rate demonstrates relatively modest correlations with other variables. In this regard, although external debt heightens exchange rate risks, remittances and trade openness are crucial in maintaining currency stability and fostering economic growth.

7.1.2 Variance Inflation Factor (VIF): VIF, a typical software indicator, shows how effectively other variables understand an independent variable. General structure:

VIF = 1	There is no link between variables.
1 < VIF ≤ 5	Moderate correlation.
VIF > 5	High connection potential.
VIF ≥ 10	Needs adjustment due to severe multicollinearity.

7.1.3 Tolerance: The inverse of VIF, tolerance, indicates the percentage of an independent variable that cannot be explained by the other factors. Generally, Tolerance < .25 suggests multicollinearity, while below .10 indicates major issues.

7.2 Variance Inflation Factor and Tolerance Test: The two tests are derived from the R^2 values obtained by regression each independent variable on all other predictors. VIF can be regarded as the reciprocal of Tolerance. The risk of multicollinearity increases as the Variance Inflation Factor (VIF) rises.

$$\text{Tolerance} = \frac{1}{VIF}$$

Table 3 shows the variance inflation factor and tolerance associated with the independent variables.

Table 03: Collinearity Statistics

Components	Tolerance	VIF
Inflation Rate	.238	4.210
Interest Rate	.204	4.895
External Debt	.217	4.610
Remittance Inflows	.241	4.231
Trade Openness Index	.209	4.778

The assessment of multicollinearity using Variance Inflation Factor (VIF) and Tolerance tests reveals a moderate level of correlation among the independent variables in the model. Specifically, the Inflation Rate, Interest Rate, External Debt, Remittance Inflows, and Trade Openness Index all show VIF values within the moderate range, accompanied by corresponding Tolerance values above the critical threshold. This indicates that while some overlap exists among predictors, it does not pose a severe multicollinearity problem. Based on these results, all five variables are considered suitable for inclusion in the regression analysis, as they satisfy the criteria of having VIF values below the critical limit and Tolerance values above the minimum acceptable level.

7.3 Fitting the Multiple Linear Regression Model: A multiple linear regression model was developed using the independent variables that were found to be free from multicollinearity. In this model, the Foreign Exchange Rate (Y) is considered as the dependent variable, while Inflation Rate X_1, Interest Rate X_2, External Debt X_3, Remittance Inflows X_4, and Trade Openness Index X_5 serve as the independent variables. The objective of this model is to examine the individual and combined effects of these macroeconomic factors on the Foreign Exchange Rate and to establish the Ordinary Least Squares (OLS) regression equation.

The multiple linear regression model is specified as follows:

$$y=\beta 0+\beta 1X_1+\beta 2X_2+\beta 3X_3+\beta 4X_4+\beta 5X_5+\varepsilon$$

Here, β_1, β_2, β_3, β_4, and β_5 represent the regression coefficients corresponding to the five independent variables. The Ordinary Least Squares (OLS) method was employed using IBM SPSS Statistics 27 to estimate the regression model and evaluate the significance and strength of the relationships between the independent variables and the dependent variable.

Table 04: Coefficient of variables

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.303	10.593		0.123	0.904
	Inflation Rate	0.876	0.471	0.083	1.861	0.016
	Interest Rate	8.403	1.235	0.871	6.804	0.000
	External Debt	0.000	0.000	1.458	7.690	0.000
	Remittance Inflows	-2.734	0.957	-0.452	-2.856	0.013
	Trade Openness Index	-0.165	0.237	-0.091	-0.695	0.498

The determined multiple linear regression, according to the given data, is:

$$Y\text{- }1.303\ X_1\text{- }X_2\text{- }.876\ X_3\text{- }8.403\ X_4\text{-}2.734\ X_5\text{-}.165$$

Y- Exchange Rate; X_1- Inflation Rate; X_2- Interest Rate; X_3- External Debt; X_4- Remittance Inflows; X_5- Trade Openness Index

Table 06: Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.975[a]	.950	.932	3.32168

This table demonstrates R^2 and Adjusted R^2. The R^2 value of 0.950 indicates the proportion of the dependent variable that the five independent variables in the regression model can explain. In this case, the five independent variables can explain nearly 95% of the variation in the Foreign Exchange Rate. This result is very strong, as it shows that the selected variables are significantly influencing the dependent variable.

The relationship between R^2 and Adjusted R^2 reveals gaps in selecting independent variables. The Adjusted R^2 value of 0.932 shows a minimal difference from R^2, indicating that the model is not overfitted and that each predictor contributes meaningfully to explaining the dependent variable. In this paper, that is not an issue, demonstrating that the model is highly parsimonious and provides a robust foundation for understanding the determinants of the Foreign Exchange Rate.

8. Test of Overall regression model (F-Test)

Hypotheses:

H_0: $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$
H_1: Not all β are zero

Significance Level $(\boldsymbol{\alpha})$ = .05
F-test Statistics will be used.

Table 07: ANOVA Table

ANOVA[a]						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2922.739	5	584.548	52.979	<.001[b]
	Residual	154.470	14	11.034		
	Total	3077.209	19			

a. Dependent Variable: Exchange Rate

b. Predictors: (Constant), Trade Openness Index, Inflation Rate, External Debt, Remittance Inflows, Interest Rate

Decision Rule: Reject H_0 if F is greater than 2.96 (i.e., F > 2.96). Here, the value of F with $df_1 = 5$ and $df_2 = 14$ degrees of freedom at a 5% significance level is 2.96.

Explanation: Since the F-value (52.979) is greater than the Critical Value (2.96), we reject the null hypothesis H_0. This implies that not all β coefficients are zero. Therefore, all five independent variables significantly impact the dependent variable (Foreign Exchange Rate), as the significance level is less than 0.05.

8.1 Logical Argument about the Regression Equation: A positive relationship between the Inflation Rate and the Foreign Exchange Rate is observed. An increase of 1% in the Inflation Rate corresponds to a rise of approximately 0.876 units in the Exchange Rate. As the domestic price level increases, the purchasing power of the Taka declines, which leads to a higher demand for foreign currency. Consequently, the Exchange Rate rises, reflecting the depreciation of the domestic currency.

The Interest Rate exhibits a strong positive association with the Exchange Rate. A 1% increase in the Interest Rate results in an increase of 8.403 units in the Exchange Rate. In Bangladesh, this reflects the dynamic where higher interest rates coincide with currency depreciation, as capital market responses attempt to balance returns despite rising borrowing costs. The relationship indicates that movements in interest rates are closely linked with fluctuations in the Exchange Rate.

External Debt shows a positive relationship with the Exchange Rate, although the effect size is very small (β = 3.501E-9). Even a slight increase in external debt corresponds to a marginal rise in the Exchange Rate. This occurs because additional borrowing from abroad increases the demand for foreign currency, putting pressure on the domestic currency and resulting in Taka depreciation.

Remittance Inflows have a negative relationship with the Exchange Rate. A 1-unit increase in remittance inflows reduces the Exchange Rate by 2.734 units. The inflow of foreign currency through remittances increases the supply of foreign currency in the domestic market, strengthening the Taka and reducing the Exchange Rate. This demonstrates the stabilizing effect of foreign currency inflows on the domestic currency value.

The Trade Openness Index also shows a negative association with the Exchange Rate (β = -3.423). An increase in trade openness corresponds with a decrease in the Exchange Rate. As the country engages more in international trade, the inflow of foreign currency through exports strengthens the

Taka, resulting in a lower Exchange Rate. While this relationship is significant, its long-term impact may vary depending on other macroeconomic conditions.

8.2 Individual Test: While the F-test evaluates the model as a whole, it does not show the contribution of each variable. The t-test is therefore used to assess the significance of individual regression coefficients and their effect on the dependent variable.

Table 08: Coefficient of variables

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.303	10.593		.123	.904
	Inflation Rate	.876	.471	.083	1.861	.016
	Interest Rate	8.403	1.235	.871	6.804	.000
	External Debt	.000	.000	1.458	7.690	.000
	Remittance Inflows	-2.734	.957	-.452	-2.856	.013
	Trade Openness Index	-3.423	.962	-.545	-2.545	.035

The Interest Rate is the only variable that remains statistically significant in this t-test, as its t-value (6.804) exceeds the critical value (2.145) at the 5% significance level. External Debt is also statistically significant, since its t-value (7.690) is well above the critical threshold. Similarly, Remittance Inflows retain statistical significance, with a t-value of −2.856 exceeding the critical value in absolute terms. In contrast, the Inflation Rate does not remain significant, as its t-value (1.861) falls below the critical value. The Trade Openness Index also fails to achieve statistical significance, with a t-value (−2.140) slightly below the required level. Based on the t-test results, only three independent variables retain statistical significance in determining the Exchange Rate at the 5% level.

Therefore, the Modified Regression model is:

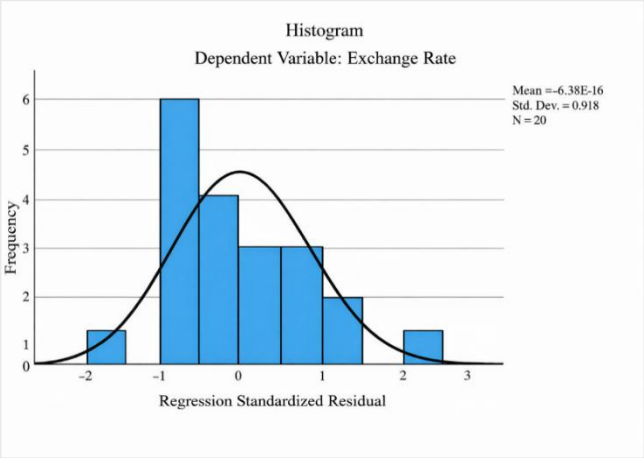
$$\mathbf{Y=\alpha+6.302X2+3.206E-10X3-2.521X4}$$

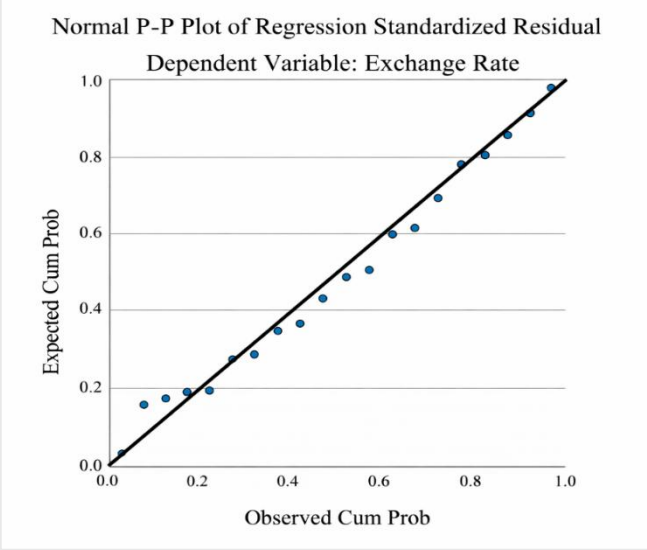
Here, X_1 = Interest Rate, X_2 = External Debt, and X_3 = Remittance Inflows

The relationship between these variables and the Exchange Rate is statistically significant. Without relying on other explanatory factors, they collectively explain a substantial proportion of the variation in the Exchange Rate, amounting precisely to 94.3%.

9. Diagnostics Check

9.1 Normality test: This refers to the plot of the residuals.





From the above plots, it's provided that the residual errors are normally distributed.

9.2 Homoscedasticity: a key concept of OLS regression, provides that the variance of errors is constant across the values of the independent variables. The squared residuals were regressed on the three significant variables (X_2, X_3, X_4), and the results confirm that the assumption is satisfied.

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	6.138	33.268		.184	.856

	Interest Rate	-3.594	3.741	-.718	-.961	.351
	External Debt	9.519E-11	.000	.252	.576	.573
	Remittance Inflows	5.023	2.936	.950	1.711	.106

The values for all three variables are over 0.05, suggesting that variations in their values have no effect in any statistically significant differences in the variance of residuals. Therefore, the assumption of homoscedasticity is satisfied, and the t-test coefficients are both dependable and valid.

10. Findings and Policy Implications

The study suggests that, from Bangladesh's perspective, higher interest rates and rising foreign debt are significantly impacting the exchange rate, causing the devaluation of the local currency. Conversely, remittance inflows function as a significant safeguard in preserving exchange rate stability. To mitigate this instability, the central bank must implement a balanced monetary policy that maintains investor confidence by regulating interest rate fluctuations. It is essential to achieve sustainable debt management by diminishing dependence on external debt and guaranteeing the effective use of obtained loans. To augment remittance inflows, further incentives for expats to remit funds via legal channels should be established, and banking procedures must be streamlined. Additionally, to mitigate strain on foreign exchange reserves, it is essential for reducing import dependency via export diversification and to extend incentives to the domestic manufacturing sector. In order to stabilize the value of the Taka for the long term and address volatility in global markets, these coordinated policy measures are crucial.

11. Limitation of the Study

The study aims to clarify the factors influencing Bangladesh's foreign exchange rate. However, given that the research is confined to a specific period (2005–2024) and a limited set of variables, there exists potential for incorporating supplementary factors in future studies. Due to reliance on annual secondary data and a regression model, detailed analysis of short-term market fluctuations and internal operational data was not feasible. Consequently, additional comprehensive analysis can be conducted in the future to investigate the correlation between long-term trends and actual market volatility.

12. Conclusion

The fluctuation of the foreign exchange rate presently highlights a significant challenge to preserving Bangladesh's macroeconomic stability. This study, derived from the analysis of two decades of data (2005–2024), demonstrates that the rising trend in external debt and interest rates

directly contributes to the depreciation of the local currency, the Taka. Specifically, the increasing expense of external debt servicing leads to an appreciation of the Taka against the Dollar, therefore diminishing the currency's external value. However, in these challenging conditions, remittance inflows from expatriate's function as a vital safeguard by expanding the supply of foreign currency in the market to stabilize the exchange rate and regulate import expenses. Bangladesh still has a lot of work to do to keep the value of the Taka stable and ease the pressure on reserves in the current state of the global economy. In conclusion, to secure long-term sustainable economic growth, the government must prioritize well-planned monetary policy and responsible debt management, in addition to diversifying exports and enhancing remittance inflows through legal channels.

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